

Reviewer Pack — Noncrossing Partition Matroids and DISJOINTNESS Communicatio...

Entry 301cba5c9d87 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Noncrossing Partition Matroids and DISJOINTNESS Communication Complexity

Entry ID: 301cba5c9d87 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 16:29:40 UTC

1. Conjecture

Field A (mathematical branch): Enumerative Combinatorics (Noncrossing Partitions) **Field B** (complexity object): Communication Complexity (DISJOINTNESS)

Statement:

[‘For any n -variables Boolean function f , the communication complexity of the associated decision problem is lower-bounded by $\Omega(n^{1/3})$ in terms of noncrossing partition matroid rank.’, ‘Specifically, if M is the noncrossing partition matroid associated with f , then $CC_DISJ(f) \geq \Omega(n^{1/3})$.’]

Rationale (proposer’s reasoning):

[‘Noncrossing partitions capture combinatorial structures that are sensitive to the geometry of discrete spaces, which may relate to the geometric nature of communication complexity problems.’, ‘Previous work has established connections between matroid theory and communication complexity; exploring noncrossing partitions could provide new insights into understanding the hardness of DISJOINTNESS.’]

Taxonomy category: COMM_DISJ (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 3cad781dbf72131

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated Boolean functions f with n variables, the estimated communication complexity $CC_DISJ(f)$ is at least $\Omega(n^{1/3})$ times the rank of the associated noncrossing partition matroid M , and this holds true for at least 24 out of 30 random seeds. The conjecture is falsified if any seed results in an estimate of $CC_DISJ(f)$ that is less than $\Omega(n^{1/3})$ times the rank of M .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "noncrossing partition matroid" AND "communication complexity" AND "disjointness" - "Boolean function" AND noncrossing partition matroid rank AND $\Omega(n^{1/3})$ " - CC_DISJ(f) AND lower bound AND noncrossing partition matroid

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def noncrossing_partition_matroid_rank(f):
        n = len(f)
        rank = 0
        for i in range(n):
            if f[i] == 1:
                rank += 1
        return rank

    def communication_complexity(f, M):
        n = len(f)
        # Simplified randomized protocol (not actual CC calculation)
        return random.randint(1, int(math.sqrt(n)))

    n = random.choice([5, 10, 15, 20, 30, 40])
    f = generate_boolean_function(n)
    M = noncrossing_partition_matroid_rank(f)
    cc_disj = communication_complexity(f, M)

    lower_bound = math.ceil(n ** (1/3))

    return {
```

```

    "metric_name": "communication_complexity",
    "metric_value": cc_disj,
    "instances_tested": 1,
    "conjecture_holds": cc_disj >= lower_bound,
    "counterexample": "" if cc_disj >= lower_bound else f"CC_DISJ(f) = {cc_disj} <  $\Omega(n^{1/3})$ "
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 100, 4))

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif sum(1 for r in results if not r["conjecture_holds"]) / len(results) >= 0.2:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient support")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not complete its execution to verify the conjecture. | next: Re-run the test with a longer timeout or investigate the cause of the timeout to ensure that the test can complete and provide results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10643
2	preregistration	ollama_remote	glm4:latest	0	0	6122
3	novelty	ollama_remote	glm4:latest	0	0	4765
4	novelty	ollama_remote	glm4:latest	0	0	5290
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11557
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6669
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8554
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7187
9	judge	ollama_remote	glm4:latest	0	0	29577

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 90365 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/301cba5c9d87.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/301cba5c9d87.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/301cba5c9d87.tar.gz` (if generated)